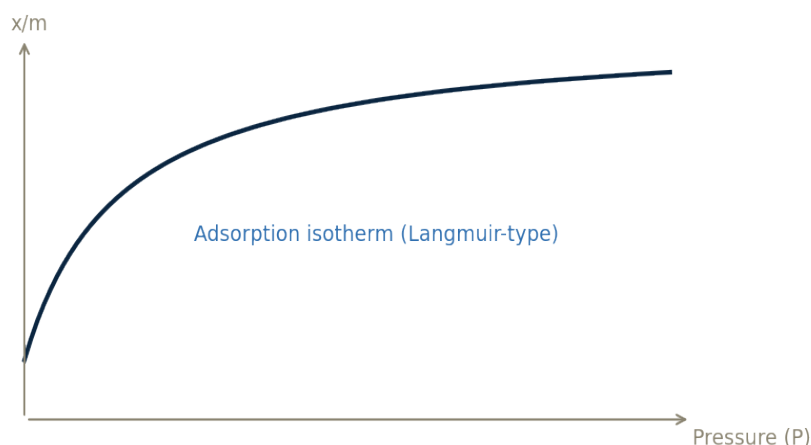


Surface Chemistry

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Adsorption is the accumulation of a substance (adsorbate) at the surface of another (adsorbent); differs from absorption, where the substance is distributed throughout the bulk.
- Physisorption (physical adsorption) involves weak van der Waals forces, is reversible, and decreases with increasing temperature. Chemisorption involves chemical bond formation, is generally irreversible, and often increases with temperature (up to a point).
- Freundlich adsorption isotherm: $x/m = k P^{(1/n)}$ (empirical relation between amount adsorbed and pressure at constant temperature).
- A catalyst increases reaction rate by adsorbing reactants onto its surface, weakening bonds and lowering activation energy (heterogeneous catalysis).
- Colloids are heterogeneous mixtures with particle size between true solutions and suspensions (1-1000 nm); examples include sols, emulsions, gels, and aerosols.
- The Tyndall effect is the scattering of light by colloidal particles, used to distinguish colloids from true solutions.
- Coagulation is the process of aggregating colloidal particles into larger particles, often induced by adding an electrolyte; the Hardy-Schulze rule states greater charge on the coagulating ion increases its coagulating power.



A Langmuir-type adsorption isotherm: amount adsorbed (x/m) vs pressure.

2. Formula Sheet

Freundlich isotherm	$x/m = k P^{(1/n)} \quad (n > 1)$
Freundlich isotherm (log form)	$\log(x/m) = \log k + (1/n) \log P$
Degree of adsorption	$x/m = \text{mass adsorbed} / \text{mass of adsorbent}$
Langmuir isotherm (form)	$x/m = (a P) / (1 + b P)$
Hardy-Schulze rule (qualitative)	Coagulating power increases with ionic charge
Gold number (qualitative)	Lower gold number = greater protective power of colloid
Tyndall effect condition	Particle size 1-1000 nm scatters visible light
Enzyme catalysis (Michaelis-Menten, qualitative)	Rate depends on substrate concentration, saturates at high [S]

3. Practice Problems (22 questions)

1. State the difference between adsorption and absorption, with one example of each.
2. State two differences between physisorption and chemisorption.
3. Explain why adsorption of a gas on a solid surface is generally exothermic.
4. State the Freundlich adsorption isotherm equation and explain what the constants k and n represent.
5. Explain how a solid catalyst speeds up a reaction using the concept of adsorption (heterogeneous catalysis).
6. Define a colloid and state the typical range of particle sizes for colloidal particles.
7. Explain the Tyndall effect and how it can be used to distinguish a colloidal solution from a true solution.
8. State the Hardy-Schulze rule and explain its significance in the coagulation of colloids.
9. Explain why physisorption decreases with an increase in temperature.
10. State one example each of a sol, an emulsion, and a gel.
11. Explain what is meant by 'protective colloid' and how gold number measures its protective power.
12. Explain why chemisorption is generally irreversible while physisorption is reversible.
13. State why finely divided catalysts are more effective than bulk catalysts, in terms of surface area.
14. Explain the term 'electrophoresis' as applied to colloidal particles.
15. State the effect of increasing pressure on the extent of gas adsorption on a solid, according to the Freundlich isotherm.
16. Explain why milk is considered a colloid (an emulsion) rather than a true solution.
17. State two applications of adsorption in everyday life or industry.

18. Explain why adding an electrolyte can cause coagulation of a colloidal solution.
19. Distinguish between a lyophilic (solvent-attracting) colloid and a lyophobic (solvent-repelling) colloid.
20. Explain why enzyme catalysis is considered a special type of surface catalysis.
21. State what happens to the Freundlich isotherm plot of $\log(x/m)$ vs $\log P$, in terms of its expected shape.
22. Explain why increasing the surface area of a solid adsorbent increases the extent of adsorption.

Answer Key

1. Conceptual - adsorption: accumulation at surface (e.g., gas on charcoal); absorption: uniform distribution throughout bulk (e.g., water in a sponge)
2. Conceptual - physisorption: weak van der Waals forces, reversible, low heat of adsorption; chemisorption: chemical bonding, irreversible, higher heat of adsorption
3. Conceptual - adsorption decreases the entropy of the gas molecules (more ordered at surface), so it must release heat (exothermic) for ΔG to be negative
4. Conceptual - k and n are empirical constants depending on the adsorbent-adsorbate system and temperature
5. Conceptual - reactants adsorb onto catalyst surface, weakening bonds and lowering activation energy, speeding up the reaction
6. Conceptual - heterogeneous mixture with particle size between 1-1000 nm
7. Conceptual - colloidal particles scatter light (Tyndall effect) making the light path visible; true solutions do not scatter light noticeably
8. Conceptual - coagulating power of an ion increases with its charge; a higher-charge ion coagulates colloids more effectively at lower concentration
9. Conceptual - physisorption is exothermic and involves weak forces; higher temperature provides molecules enough energy to escape the surface, reducing adsorption
10. Conceptual - sol: e.g., gold sol; emulsion: e.g., milk; gel: e.g., jelly/agar
11. Conceptual - protective colloid prevents coagulation of another colloid; lower gold number means less protective colloid is needed, so higher protective power
12. Conceptual - chemisorption involves strong chemical bonds requiring significant energy to break, making it irreversible; physisorption involves weak forces, easily reversed
13. Conceptual - finely divided catalysts have much greater surface area per unit mass, providing more active sites for adsorption and reaction
14. Conceptual - electrophoresis is the movement of charged colloidal particles under an applied electric field, toward the oppositely charged electrode
15. Conceptual - extent of adsorption increases with pressure but the rate of increase slows at higher pressure (non-linear relation)
16. Conceptual - milk consists of fat droplets dispersed in water (particle size in colloidal range), scattering light and showing other colloidal properties
17. Conceptual - e.g., use of activated charcoal in gas masks; use of adsorption in heterogeneous catalysis (e.g., Haber process)
18. Conceptual - electrolyte ions neutralize the charge on colloidal particles, reducing repulsion and allowing particles to aggregate (coagulate)
19. Conceptual - lyophilic colloids have strong affinity for the solvent (e.g., starch in water, stable); lyophobic colloids have weak affinity (e.g., metal sols, less stable, need stabilizers)
20. Conceptual - enzyme (catalyst) surface adsorbs substrate at active sites, similar to solid heterogeneous catalysis, before releasing product
21. Conceptual - expected to be a straight line, with slope $1/n$ and intercept $\log k$
22. Conceptual - more surface area provides more active sites for adsorbate molecules to attach, increasing total adsorption

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.